Raj Gajjar

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Summary

Full-stack AI engineer who ships end to end: agent systems, backend APIs, and frontend. Built a real-time voice agent on LiveKit with tool-calling and multi-provider LLM fallback, an autonomous coding agent with sandboxed execution, and a production AI proposal-generation platform with e-signature integration. Open source contributor in OWASP and Layer5.

Education

LJ Institute of Technology Ahmedabad, India
Bachelor of Engineering in Computer Science and Engineering *September 2023 – August 2027*

Experience

Associate AI Engineer July 2026 – Present
E2M Solutions, Ahmedabad, India

- Building production LLM applications with LangChain and LangGraph for client-facing AI products, including an automated lead generation platform and an end-to-end AI proposal generation system with contract drafting and e-signature workflows

AI/ML Engineering Intern January 2026 – June 2026
E2M Solutions, Ahmedabad, India

- Developed custom AI workflows and LLM-powered automation tools supporting client-facing AI products

Full Stack Developer Intern December 2025
Asambhav Solutions, Ahmedabad, India

- Built AutoLead, a RAG-based conversational agent using LangChain and LangGraph with a Django REST Framework backend, enabling natural language booking of calls, test drives, and service appointments with semantic search over pgvector embeddings

Projects

Voice Agent: Real-Time Voice Assistant (LiveKit, FastAPI, React) github

- Built a real-time voice assistant on LiveKit that manages tasks and books Google Calendar events through live voice conversation, with Google OAuth and full conversation history
- Architected a three-service system (FastAPI backend, React frontend, LiveKit agent worker) with Deepgram speech-to-text, Groq LLM with automatic Cloudflare Workers AI fallback, and ElevenLabs text-to-speech
- Designed a tool-calling agent for task and calendar management with fuzzy matching and double-booking prevention, backed by Postgres and Redis, deployed as independently scaled services on Railway

Sage: Autonomous Coding Agent (Python, FastAPI, SQLite, Docker) github

- Built an autonomous CLI coding agent that explores a codebase, generates an execution plan, and executes it step by step after mandatory user approval, using a Plan Mode and Execute Mode workflow
- Implemented a workspace sandbox confining all file and shell operations to a configured directory, enforced at the point of execution
- Engineered multi-provider LLM support (Groq, ZhipuAI) with automatic round-robin fallback on rate limits, resumable SQLite-backed sessions, and a versioned FastAPI service with Docker Compose deployment; maintained quality with pytest, ruff, and mypy

Smart Inventory Management: AI-Powered Inventory Optimization github

- Developed an intelligent inventory system to reduce food waste by predicting expiry dates and generating personalized recipes
- Integrated the Gemini API for AI-generated recipes based on available ingredients and trained a custom YOLO model for real-time freshness detection

Open Source Contributions

- **OWASP Agent Security Regression Harness**: 3 merged pull requests: GitHub Actions CI example for security regression testing, pytest-cov coverage gate at 80%, and AI-assisted contribution policy documentation
- **Layer5 Sistent** (React design system): 2 merged pull requests extending the open source icon library
- **TracerCloud SSRE Agent**: merged pull requests adding test cases for the agent

Technical Skills

Programming Languages: Python, JavaScript, Java, C++, SQL
Web Development: FastAPI, Django, Django REST Framework, React.js, Next.js, Node.js, Express.js, Astro.js, REST APIs, WebSockets, OAuth 2.0
AI and Machine Learning: LangChain, LangGraph, LangSmith, LiveKit, Retrieval-Augmented Generation (RAG), Tool-Calling Agents, NumPy, Pandas, Scikit-learn
Databases: PostgreSQL, pgvector, MongoDB, MySQL, SQLite, Redis, Pinecone
Tools and Platforms: Git, GitHub, GitHub Actions, Docker, Linux, Railway, Supabase, Tailwind CSS, shadcn/ui, AWS, Cloudflare